Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

Haben Michael[1]

[1]University of Massachusetts, Amherst, MA (hmichael@math.umass.edu)

Abstract. stijnen describes 4 random effects models for meta-analyses of involving event outcomes. inference for these models is based on an asymptotic chi-squared approximation. we describe a procedure for bartlett correction of the confidence interval, improving the convergence rate from $1/n$ to $1/n^2$ under ideal conditions. we apply the proposed procedure to the meta-analysis considered by stijnen 2010, finding ... . Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1 Introduction

There is currently underway a move to favor generalized linear mixed models in carrying out meta-analyses of event outcomes, particularly in when sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions, such as gaussianity of the effects and perfect knowledge of standard deviations, required by the traditional alternative, inverse-variance weighting. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are suspect, though smaller meta-analyses are where the use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modelling assumptions to improve the efficiency of the testing procedure. When the modelling assumptions are met, the improvement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$. [[the key modelling assumption in this setting is the gaussianity of the random effect.]]

While Bartlett corrections have long been applied to generalized linear models [[cite]], there do not appear to be extensions to even the simple GLMMs that are used in meta-analysis. The reason for the reluctance appears to be computational difficulty [[cite hanada]]. While computer algebra systems have long been applied to compute out the third- and fourth-order derivative arrays needed for Bartlett correction of GLMs ((cite old cordeiro paper, andrews and stafford, mccullagh maybe)), closed-form expressions for GLMM Bartlett corrections are unavailable for any of the commonly used models other than the normal-normal/identity link ((cite noma, michael2026)). A combination of automatic symbolic manipulation and numerical approximation appears necessary. We argue that with careful design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we summarize four event data meta-analysis models, following [[Stijnen's]] exposition. In Section 3 we describe a practical approach to estimation of the Bartlett correction applicfable to these four models. In Section 4 we examine the effectiveness and robustness of the corrected test, finding ... . In Section 5 we re-visit the data analyses of ((stijnen)), finding ... .

2 Background

2.1 Statistical models

We consider the random effects model for a meta-analysis consisting of n studies. In this model, a Gaussian random variable models the effects $\theta_1, \ldots, \theta_n$, of the individual studies in the meta-analysis,

$$\theta_i = \mu_0 + \tau_0 Z, \qquad Z \sim N(0,1)$$

The target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. The distribution of the data given the random effect depends on the type of analysis being conducted. Following ((stijnen)) we consider four types of analyses, determined by whether the studies included are single-arm or two-arm trials, and whether the estimand is based on proportions or incidence.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$f_{m_i}(y_i \mid Z_i) = \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i}$$
$$p = p(\mu_0, \tau_0^2) = \text{expit}(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as a parameter of the conditional PMF f_{m_i}, which therefore typically varies across the studies in the meta-analysis.

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The statistical model is conditional on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, under which $Y_i^{(T)}$ follows Fisher's non-central hypergeometric distribution

$$f_{m_i^{(T)}, m_i^{(C)}, S_i}(y_i^{(T)} \mid Z_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}}\omega^{y_i^{(T)}}}{\sum_{\max(0, S_i - m_i^{(C)})}^{\min(S_i, m_i^{(T)})} \binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i - y}\omega^{y}}$$
$$\omega = \omega(\mu_0, \tau_0^2) = \exp(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the ratio of the treatment to control log-odds. The parameters of the conditional PMF are $m_i^{(T)}, m_i^{(C)}$, and S_i.

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f_{T_i}(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!}$$
$$\lambda = \lambda(\mu_0, \tau_0^2) = T_i \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the log of the incidence rate. The follow-up time T_i is treated as a parameter of the conditional PMF.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. Conditionally on $S_i = Y_i^{(T)} + Y_i^{(C)}$, $Y_i^{(T)}$ is binomial

$$f_{T_i^{(T)},T_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \binom{S_i}{y_i^{(T)}} p^{y_i^{(T)}}(1-p)^{S_i - y_i^{(T)}}$$
$$p = p(\mu_0, \tau_0^2) = \text{expit}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

The target of inference is the logarithm of the ratio of the incidence rates between the two arms. The sum of events S_i and and the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as parameters of the conditional PMF.

In all cases the likelihood for study i has the form:

$$L_i(\mu, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f_i(Y_i|z)\phi(z)dz, \qquad i = 1, \ldots n$$

where $f_i(y|z)$ is study i's PMF conditional on a Gaussian random effect, given above. Furthermore, the model parameters (μ, τ^2) only enter the likelihood ((ref)) through a parameter of the conditional PMF, the mean λ in the case of the Poisson, the success probability p in the cases of the binomial, or the odds ratio ω in the case of the non-central hypergeometric.[[maybe put this up]]

The likelihood ratio test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2\sum_{i=1}^{n}(\log L_i(\hat{\mu}, \hat{\tau}^2|Y_i) - \log L_i(\mu_0, \hat{\tau}_0^2|Y_i))$$

where $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto \sum_{i=1}^n \log L_i(\mu, \tau^2|Y_i)$ and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu)$ maximizes $\tau^2 \mapsto \sum_{i=1}^n \log L_i(\mu_0, \tau^2|Y_i)$. In routine applications W converges to a chi-squared distribution under the null hypothesis, by Wilks' Theorem. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.

2.2 Bartlett correction

Under a range of conditions [[citations]] the mean of a correctly specified LR statistic ((ref)) admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

Therefore, the Bartlett-corrected statistic

$$W_B = \frac{W}{1 + b/n}$$

has a mean agreeing with that of a chi-squared random variable up to order $O(1/n^2)$. The work of ((lawley and bartlett)) shows further that, under appropriate conditions, all other moments, and so also the entire distribution, also agree with a chi-squared up to order $O(1/n^2)$. As with ((ref lr statistic)), the adjusted statistic may be inverted to obtain CIs and p-values. [[mention b depends on parameters, or maybe make explicit in the notation, and that 1/n2 rate can be achieved when these are estimated by root-n consistent estiamtros such as the mle's.]]

lawley ((cite)) provides a formula for the correction ((ref)) as sums and products of the components of the inverse Fisher information for ((ref likelihood)) and certain tensors [[k terms]]. Being a small set of sums and products this formula does not introduce numerical or computational difficulties that aren't already present in the basic tensor components, and we focus here on the basic components. To this end, let D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. For example, $D_{\mu,\tau^2} \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ denotes the off-diagonal term of the observed information matrix for ((ref total likelihood)). Define $k_r = k_r^{(0)} = \mathbb{E} D_r \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ and $k_r^{(s)} = D_s k_r$. In this notation, for example, the Fisher information matrix has components $k_{\mu,\mu}, k_{\mu,\tau^2}, k_{\tau^2,\tau^2}$. Lawley's formula for the Bartlett correction requires all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s of length 1 or r is of length 2 and s of length 1 or 2. Taking into account permutation symmetry of the order of differentiation, there are 35 such terms in a model with one parameter of interest and one nuisance paramter.

By linearity of expectation and differentiation, it suffices to compute the terms $k_r^{(s)}$ for the unit log-likelihoods $\log L_i(\mu, \tau^2 \mid Y_i), i = 1, \ldots, n$. Let us consider a single study from the meta-analysis, dropping the subscript i from the conditional PMFs and likelihoods. Then the general tensor component needed for the computation of Bartlett's correction is

$$k_r^{(s)} = \sum_y D_s \left(\left(D_r \log \int f(y|z)\phi(z)dz \right) \int w_k f(y|z)\phi(z)dz \right)$$

The outer sum is taken over the support of Y.

2.3 Assumptions

Among the assumptions underlying the above development we focus on two prominent ones. The first is correct specification of the likelihood, in particular, gaussianity of the unobserved random effect. [[we refer to sitjnen to other modeling assumptions implicit in ((ref 1–4 above)), such as ... .]] While the LR test itself depends on gaussianity for valid inference ((citation showing effects of bad estimatio of tau2)), the Bartlett correction doubles down on the assumption. Second, the arguments under which the Bartlett correction delivers a $1/n^2$ rate are usually justified by Cramér's condition for the underlying random variables. This condition excludes lattice variables, such as are considered here. Consequently, while Bartlett corrections have been derived for statistics based on lattice random variables since at least

the 80s, their effectiveness does not rest on a firm theoretical foundation and in fact does not appear to be understood in much generality ((cite frydenberg re multinomial ngative result and gotze, and janson re positive result)). We have no theoretical contibutions to make here and will examine both model robustness and the effect of discreteness by simulation.

[[add a third issue. say at outset 2 issues are really issues with the lr test generally, and the 3rd is about the bartlett correction. wilks theorem applies to interior points. when tau2=0 ie fixed effects model, not applicable. some simulations have suggested does not affect inference for mu, but difficult to see this can be so in general, since inference for tau2 is known to be off ((self,liang)) at the boundary and mu and tau2 are not orthogonal in our models. means for lr test must check tau2 interior is reasonable eg examining the mle, or perhaps computing a fixed effects compute the fixed effects LR CI. [[with iv method can think of re model as a generalization of the fe, if the data is actually fixed effects CIs are valid there is just a loss of power. in the case of the lr, the random effects procedure is different from the fixed effects procedure.]]

3 Computing the tensor components ((k))

3.1 Quadrature

To compute the tensor component ((ref)), we approximate the integrals by a quadrature rule. Since all the likelihoods ((ref)) involve integrating against a Gaussian, Gauss-Hermite quadrature is a natural choice. This and similar methods ((cite biometrics paper)) provide weights $w_1, \ldots, w_{n_{GH}}$ and nodes $z_1, \ldots, z_{n_{GH}}$ in the domain of integration allowing the Gaussian integral of smooth integrands such as appear in ((ref)) to be approximated by a weighted sum,

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|z_j) \right) \sum_k w_k f(y|z_k) \right)$$

The number of nodes n_{GH} controls the quality of the approximation and introduces a tuning parameter into the proposed method. A further tuning parameter is whether fixed or adaptive GH is to be used. fixed GH uses the same choice of nodes irrespective of the integrand, eg always centered at 0, namely, the roots of ... . on the other hand, adaptive GH transforms the nodes to take into account of the mode and scale of the integrand. fixed appears fine in our experience when the pmf has a support concentrated not too far from 0. however for the poisson model above, with unbounded support, and possibly for the other models ((ref)) when the support is large though bounded, adaptive may be preferable. [[maybe give poisson example in data anlaysis, and cite to it]]. the increase in computation cost to compute the mode and support may be offset by the reduced number of nodes to obtain a comparable approximation to fixed GH. The main difficulty with adaptive is that the mode and scale of the integrand, and therefore also the AGH weights and nodes, typically depend on the parameters mu,tau2, complicating the partial differentiation.

intremediate approach: we ignore this dependence, leading to an approximation intermediate in quality beween fixed and adaptive GH. this means that using adaptive over fixed GH is advantageous only when the mode and scfale of hte likelihood doesnt change to much

under differentiation. when we do so, however, the cost of adaptive GH becomes very cheap. the only integrand for which adaptive GH needs to be computed are o fthe form condpmf phi. as all the cond pmfs belong to the exponential family, these are strictly log-concave and unimodal, and the mode maybe obtained efficiently using Newton's method. with mode given by the solution to: and scale by: .

3.2 Differentiation

Carrying the differentiations ((ref above display)) through yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$, say, $F_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k}\frac{d^l}{d(\tau^2)^l} f(y|z_j)$. These expressions for ((k)) in terms of the F_{kl} are the same for any model based on a likelihood of the form ((ref)) and a quadrature approximation, and may be pre-computed. For example $k_{\mu,\mu} = \sum_y (F_{20}(y) - (F_{10}(y)^2/F_{00}(y)))$, and $k_{\mu,\tau^2}^{(\tau^2)} = \sum_y (F_{01}(y)^2 F_{10}(y)/F_{00}(y)^2 - (F_{02}(y)F_{10}(y) + F_{01}(y)F_{11}(y))/F_{00}(y) + F_{12}(y))$.

The derivatives $F_{jk}(y)$ of the model-specific conditional PMFs that appear in the 35 tensor components number 15: $F_{jk}(y)$ for $0 \le j + k \le 4$. The form of these derivatives may be derived independently of data but their specific values in general depend on parameters of the conditional PMF ((see above)), and must typically be computed for each study of a meta-analysis. The latter computations may be carried out in parallel when the meta-analysis consists of stochatically independent studies, such as we consider here. Furthermore, since the conditional PMFs only depend on the model parameters (μ, τ^2) through the PMF's distributional parameters, the derivatives $F_{jk}(y), 0 \le j + k \le 4$, may be re-written as linear combinations [[linear for binomial and hyperg also??]] of $F_{00}(y) = \sum_j w_j f(y|\theta_j)$ using standard recurrences ((cite johnson and kotz univariate discrete distributions)). The advantage of this form is that the terms F_{00} involve only the PMF itself, for which optimized and numerically stable routines are available. For example, two applications of the Poisson PMF recurrence $\frac{d}{d\lambda} f_\lambda(y \mid Z) = f_\lambda(y-1 \mid Z) - f_\lambda(y \mid Z), y \ge 1$, yield, for model ((ref poisson/incidence rate model)),

$$\begin{aligned} F_{11}(y) &= \sum_j w_j \frac{d^2}{d\mu d\tau^2} f_{\lambda(\mu,\tau^2)}(y \mid z_j) \\ &= \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y-2) + (\frac{d^2\lambda}{d\mu d\tau^2} - 2\frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2}) F_{00}(y-1) + \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y), \qquad y \ge 2 \end{aligned}$$

In summary. the approach is, for each of the n studies in the meta-analysis: evaluate the 15 terms $F_{jk}(y)$ [[for y in support for the ith study, or truncated in case of poisson]] using only routines for the conditional PMF (Poisson, binomial, or hypergeometric) and basic arithmetic, and then compute the ((35)) k terms using only the computed 15 ((F)) terms and basic arithmetic.

3.3 Tuning parameters

1. fixed vs adaptive
2. n.GH, poisson truncation point. follow usual approach with profile based LR, where examination of the profile curve guides.

3. mu grid. for the computation of a CI based on the profile likelihood, calculations described in ((ref differentiatoin section)) must then be repeated over a grid of μ values.

4 Simulation

we examine by simulation the rate at which the LR and Bartlett-corrected LR statistics converge to the reference chi-squared distribution under the four models ((ref)). The purpose is three-fold: 1. to investigate whether the quadrature approximations used by the proposed method are sufficient, 2. to investigate whether the bartlett correction is effective as applied to this type of lattice data, as there are few theoretical results here, and 3. to assess the robustness the of any improvement in the rate offered by the corrected statistic to deviations from the assumed Gaussian model for the random effect.[[add tau2 boundary issue here]]

4.1 Design

to make the simulation results relevant, the synthetic meta-analyses are constructed to resemble the data of the stijnen tables ((data analysis)). the true munull and tausqnull are taken to be the mle's for the data obtained by stijnen et al. the distributional parameters are obtained by sampling with replacement from the stijnen tables. e.g., for the control-arm ((bn-prop)) model, a meta-analysis of size n is sampled by 1. sampling with replacement n trial sizes $m^{(C)}$ from the 18 control arm study sizes in ((ref nw18 data set)), 2. sampling n effects Y under the binomial likelihood ((ref)) with the sampled trial sizes and the parameters $(\mu_0, \tau_0^2) = (...)$ taken from the MLE calculations of ((stijnen2010)).[[e.g., number of trials mi in the ((bin-prop model)) or total number of events s in the ((hyperg or bn-irr models))]] meta-analyses of sizes $5, \ldots, 15$, are sampled in this manner, and the rate of convergence of the observed kolmogorov-smirnov distanc to a chi-squared is estimated. ((see other ma paper for details on rate estimation)).

robustness of the gaussianity assumption is examined by perturbing a normal to a skewed distribution ((cite pusey-jones paper)). the skew was chosen since it drives the first-order error in CLT-type approximations.

((add mle approximation info. concession to hardware limitation. like using n4 rather than n5 mc reps.)) the practical reason is to avoid the prohibitive computational cost of recalculating the bartlett calculation for each of the n^4 monte carlo replicates. the justification are the results of [[barndorff and cox]] that plugging in root-n consistent eestimators into the bartlett correction term b(mu,tau2) only introduces an O(1/n2) error.

details.(put in appendix? or maybe just cite to other ma sim section?) rate of convergence is estimated as follows. For sizes $n = 5, \ldots, 15$, $B = n^4$ meta-analysis datasets of size n are sampled, and the ((LR)) and ((LR-bartlett)) statitsics are computed. maximum distance of the empirical CDFs of the B ((LR)) and B ((LR-bartlett)) statistics to a chi-squared distribution with df=1 is then recorded. ((actually prob integral transformation is applied, then distnace to uniform. should be the same.)) the rate of convergence of this sequence of max distances to 0 is then approximated by the slope coefficient from regressing their logs against the log of n. With $B = n^4$ ((need to introduce B notation?)) replicates, the monte

carlo error of the empirical CDFs is expected to be $O(1/n^2)$, which is the same order as the ideal bartlett-corrected rate of convergence. this rate estimate is then sharpened by taking the median across 100 replicates under this process. [[this approach is taken as a concession to the computational cost of forming $B > n^4$ monte carlo repetitions.]] [[repeating this process of obtaining a rate estimate is then repeated 100 times, with the median reported in ((the figures. or if using a table, get mean and sd too)).]]

4.2 Results

5 Data analysis

We revisit the example data analyses of ((stijnen2010)). Meta-analyses are carried out under the models ((ref)) and 95% confidence intervals for the underlying effect μ_0 are calculated using the corrected and uncorrected profile LR. The results are presented in Table 1.

5.1 Proportions

((niel-wiese 2007)) carries out a meta-analysis on the efficacy of an anti-infection treatment for catheters. Each of the 18 included studies reports, for treatment and control arms, the total number of patients in the study and number of those who developed a catheter-related infection ((ref Table. should mention this incorporates the corrections mentioned by metafor guy.)).

We apply the binomial-prpo model ((ref)) to obtain CIs for the log odds of infection in each arm separately and the hyperg model ((ref)) to obtain a CI for the log-odd ratio across the arms. In all three cases, the CI using the bartlett corrected profile likelihood is only slightly wider than the uncorrected, with the endpoints differing only in the hundredth's place or less. This agreement may reflect the moderate sample size of the the meta-analysis at 17–18 studies.[[effectively 17 for hn model due to double zero study]].

[[give sample plot]]

5.2 Incidence

niel-weise ((2008)) also carries out a meta-analysis on the efficiacy of an anti-infection treatment for catheters. Each of the 9 included studies reports, for each arm, the number of catheter-related infections observed in a population as well as the total "catheter-days", the sum over the population of the number of days each subject used a catheter.

We use the poisson-normal model ((ref)) to obtain CIs for the log incidence rate in each arm, and the binomial-irr model ((ref)) to obtain a CI for the log-incidence-rate ratio. For this smaller meta-analysis the difference between the corrected and uncorrected CIs is more dramatic.

[[give sample plot. maybe give poisson model with adaptive and non-adaptive GH. cite latter in method/tuning parameter section.]]

Estimand	**Meta-analysis**	**LR CI**	**Proposed CI**	**% Diff**
log odds	stijnen-1, control arm	[-3.93, -2.92]	[-3.93, -2.90]	+2.2%
	stijnen-1, treatment arm	[-5.57, -4.13]	[-5.58, -4.12]	+1.7%
log-odds ratio	stijnen-1	[-2.21, -0.71]	[-2.25, -0.68]	+5.1%
log IR	stijnen-6, control arm	[0.90, 1.90]	[0.85, 1.95]	+10.6%
	stijnen-6, treatment arm	[0.10, 1.55]	[0.03, 1.62]	+9.8%
log-IR ratio	stijnen-6	[-1.05, -0.01]	[-1.12, 0.05]	+11.3%

Table 1: Standard and Bartlett-corrected confidence intervals for the example meta-analysis datasets considered by ((stijnen)). There is little advantage in Bartlett correction for the meta-analyses [[nw87 data]], which consists of 18 studies. There is a greater difference for the meta-analyses based on [[nw88 data]], which has half as many studies, including a loss of significance at the nominal level $\alpha = .05$ for the log-IR ratio study.[[switch first and second columns]]

6 Conclusion

many of the required assumptions for the proposed method are assumptions of the LR. eg non gaussianity of the RE affects inference for the LR as well. ideal would be if the correction could always be used, would provide a boost when the assumptions are met but not harming inference when not met.